		$v_{max} = 9.3 \times 10^5$ m s^{-1}
30	**D**	Rate of decay $A = \lambda N = \frac{\ln 2}{t_{1/2}} N = \frac{\ln 2}{t_{1/2}}(nN_A)$ where n and N_A are number of moles and Avogadro's number respectively. Hence $A \propto \frac{n}{t_{1/2}}$. The ratio $\frac{200}{6200}$ is the smallest, i.e. 0.0323. Option A is 0.0476 Option B is 0.0755 Option C is 0.129